Name:______________________________ ( ) Class:________

Overview of Practical QA Skills:

- Read, predict (if possible), carry out and write best possible answer that is relevant and concise.
- Use the notes to help work out unknown or explain ideas.

Note: Need to revise the following topics - acids, alkali, cations, anions, gases, metals (displacement), redox, decomposition of solids, PRECIPITATION, solubility of salts.

PRACTICAL TECHNIQUES (general guidelines, please always follow the instructions given):

Analysing substances to find out what they contain and their nature, is an important skill.

(a) identify unknown salt
(1) (if solid) dissolve small amount of salt in distilled water to make a solution (filter to obtain solution if necessary)
(2) (if solution) – test for cations and anions

(b) use correct amount
- solid; enough to cover bottom hemisphere
- liquid/solution; approximately 1 cm depth for each test
(do not take too much sample for testing as ppt formed might be more than you can dissolve)
*if you suspect the ppt is actually soluble but you are unable to fully dissolve the ppt due to too much ppt being formed in the test tube, pour out the ppt leaving behind enough to cover the bottom hemisphere, then add the excess reagent
- limewater; approximately 2 cm depth to test for CO_2 (for the delivery tube method)
approximately 5 cm depth (or enough volume for the teat pipette to dip in) to test for CO_2 (for the teat pipette method)

(c) use distilled water
- to make solutions
- to wash residue
- to rinse apparatus used for titrations

(d) use solutions/reagents
- **shake solutions before use**
- add excess reagents only if asked to do so or asked to add until no further change.
- add reagents slowly without spillage
- ensure good mixing by swirling/stirring
- leave test tube to stand for some time before washing off

(e) heat solutions
- tilt test tube at angle of 45°
- shake solutions from time to time
- remove from flame if solution looks as if it will boil over
- point mouth of test tube away from yourself & others

(f) follow instructions
- e.g. warm instead of heat
- e.g. acidify reagents such as potassium manganate(VII) etc.

(g) test for gases
(i) litmus paper
- must be damp using distilled water
- to be held over mouth of test tube (without touching)

(ii) burning splint
- to be inserted into the mouth of the test tube

(iii) glowing splint
- to be pushed into test tube (about 1 to 2 cm above reacting mixture) (avoid touching sides of test tube which may be damp)

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(iv) delivery tube - fit properly, with the longer tube inserted into the limewater.
- if heating involved, remove the tube that is inserted into the test tube of limewater before stopping the heating.

(v) teat pipette - can be used as alternative to test for carbon dioxide
- prepare sufficient limewater (about 5 cm depth) in another test tube for teat pipette to reach.
- press teat pipette, lower it into test tube above solid being heated & let go to draw in gas.
- bubble gas into limewater by pressing teat pipette
- repeat the previous two steps until the first formation of white precipitate is seen.

(vi) Which gas to test 1st? – look out for clues in the steps to pre-prepare test for gas (refer to part (n))
- if no clues, use litmus paper as:
  - ammonia is alkaline
  - carbon dioxide and sulfur dioxide* is acidic
  - hydrogen and oxygen are neutral
  - chlorine will bleach litmus paper

*no longer tested in practical, for your information only.

(h) handle precipitate
- allow reaction mixture to stand for a few minutes to identify colour of ppt & solution
- tilt test tube (or filter) to see colour of ppt (if necessary)
- need only little ppt to test for its solubility

*(i) using concentrated acid
- use test tube holder for test tube, collecting it from fume cupboard
- be careful when washing

*last appeared in 1994

(j) record observations
- use appropriate terms:
  1. powder / crystal = appearance of solid
  2. turned = change of colour
  3. melted = change of state
  4. liquid = obtained after powder / crystals melt
  5. dissolved = solid disappears in liquid / solution
  6. solution = obtained after solid dissolves
     (The term "mixture" is rarely applicable for QA.)
  7. residue = solid left after heating / filtration
  8. ppt = solid obtained after **<u>mixing 2 aqueous solutions</u>**
  9. deposits = solid formed from displacement reactions or electrolysis
  10. pale yellow / yellow / dark yellow / bright yellow = different shades of yellow
- check if test tube feels warm / hot (i.e. heat energy is given out)
- correspond with respective parts of each test, showing clear alignment
- look out for gas produced (e.g. effervescence / colour / ~~odour~~)
- write observations for gases and name the gas (refer to QA notes given)

e.g. Gas forms white precipitate in limewater. Gas produced is carbon dioxide.

(k) heat solids

- use clean / **DRY** test tube
- heat gently horizontally, then steadily increase heating until no further change (when heating strongly, air hole must remain fully opened)
- test for gas produced
  (i) think of which test to do
  (but usually there are hints given. e.g. prepare litmus paper or limewater)
  (ii) test for gas
  (iii) may need to repeat
- complete heating even after identifying gas
- note colour change / change of state of solid
- note colour of residue when hot and when cold
- write observations for heating of solids as follows:

(1) **What happens on heating?** e.g. **On heating**, white powder/solid turns yellow (i.e. There is ONLY a change of colour. Need to write the colour before heating and after.) OR **On heating**, brown powder melts to give a red liquid. (i.e There is a change of state. Need to write the colour before and after change of state.)
(2) **Which gas is given off?** e.g. **Gas** relights glowing splint. Gas produced is oxygen.
(3) **Were there droplets of colourless liquid found on sides of test tube?** - applicable for heating of solids, not for heating solutions/liquids (may not appear) - **DO NOT** write as "droplets of ~~water~~ is formed" (observations is not conclusion)
(4) **What happens on cooling?** e.g. **On cooling**, yellow powder / solid turns back to white. (Can be same colour, e.g "white solid remains white upon cooling") OR **On cooling**, a yellow residue is obtained.

(l) pre-empt observations, getting ready necessary (e.g. prepare test for gas before carrying out the steps) (Refer to "Suggested Approach in Handling QA Practical")

(m) draw conclusions

Note: If the conclusion is correct, but the observation is not written down. The correct conclusion would be marked as wrong. The conclusion must be supported by observations and not through guessing.

Please read carefully what the question wants.

e.g. (1) **Name the metal / salt / acid** present.	copper / copper(II) sulfate / sulfuric acid
(2) **Name the metal ion / cation / anion** present	copper(II) ion / hydrogen ion / sulfate ion
(3) Give the formula(e) of the **metal ion / acid** present	Cu^{2+} / H_2SO_4
(4) Unknown ion:	chloride ion
Evidence:	A **white precipitate** is formed with **acidified silver nitrate (aq) in Test 2(a)**.

(n) consider the following clues to predict for gases (to prepare test for gas before seeing effervescence):

S/N	Procedure (Type of Reaction)	Identity of Unknown	Look out for
1	Heat a sample of powder (decomposition) (usually in the steps, they will tell you which gas to prepare for)	NH_4^+	ammonia
		CO_3^{2-}	carbon dioxide
		NO_3^-	brown gas (NO_2)* / oxygen *NO_2 is acidic.
		IO_3^{2-}	violet vapour (I_2) / oxygen
		SO_3^{2-}	sulfur dioxide** **will not be tested as hands on practical, only for theory
2	Add grey / silver **powder** to an acid (reaction between reactive metal + acid)	metal	• hydrogen gas (effervescence) *may not produce enough volume to extinguish with a pop sound • powder dissolve to give a …………… solution
3	Add an acid to coloured **powder** (reaction between carbonate + acid)	CO_3^{2-} / HCO_3^-	• carbon dioxide gas (effervescence) • powder dissolve to give a …………… solution
4	Warm white **powder** with an alkali (reaction between ammonium salt + alkali)	NH_4^+	ammonia
5	Warm ammonium chloride with a **solution** (reaction between ammonium salt + alkali)	alkali	ammonia
6	Warm **solution** with sodium hydroxide / with sodium hydroxide + aluminium foil gently	NH_4^+ NO_3^-	ammonia
7	Add hydrogen peroxide / potassium manganate(VII) / potassium iodide	-	oxygen
8	Add grey / silver **powder** to a salt solution (displacement reaction / or metal + water)	metal	• reddish-brown / grey / silver despositED • colour change in salt solution • may have effervescence (hydrogen gas)

Suggested approach in handling QA practical

1. Take a deep breathe.

2. Look through the pages, so you know how fast you need to do the experiment.

3. Read the opening lines, there could be important information.

(e.g Is it a salt? 3 ions? Metal salts? Acidic? ..etc)

4. For any possible predictable observations from the steps, be ready for possible observations. (Revise some theory before practical exam!)

- Cation – When question says add <u>aqueous sodium hydroxide/aqueous ammonia</u>

What is the colour of precipitate formed? Is it soluble or insoluble in excess?

Note: If no precipitate, it might be H^+ or Group 1 ions!

e.g. Add 2 cm^3 of <u>sodium hydroxide</u> into Solution B, then till no further change.

<u>(Add a piece of aluminium foil) Gently heat/warm</u> the mixture.....

Once you add aqueous sodium hydroxide, followed by warm → always test for ammonia (prepare damp red litmus paper!)

- If ammonia is produced and the step has no Al foil → ammonium ion present
- If ammonia is produced and the step has Al foil → nitrate ion is present.

- Anions – carbonate, sulfate, <u>chloride, iodide</u>, nitrate

steps says acid added to solid (NOT Metals) and bubbles are seen being produced.

Check using limewater!

steps contains "acidified barium nitrate/barium chloride"

→ expect white precipitate or no precipitate / no visible reaction (negative test)

steps contains "acidified silver nitrate"

→ expect white precipitate (Cl^- ion), yellow precipitate (I^- ion) or no precipitate / no visible reaction (negative test)

- Other reactions good to review: Acid/alkali reactions, displacements, metal + acid, redox, organic, solubility table, preparation of salts

any steps that has metals reaction and produces bubbles → prepare hydrogen test, lighted splint!

any substance that has purple acidified potassium manganate(VII) change colour to colourless → that substance is a reducing agent, itself got oxidised any substance that has colourless aqueous potassium iodide change colour to brown → that substance is an oxidising agent, itself got reduced

Technique reminder: When adding the oxidising and reducing agent, **the recording of observations should be focused on the oxidising and reducing agent colour change**, rather than the final colour of the mixture.

- Know how various gases are formed (get ready suitable test for gases, only unknown gases then use litmus),

only for heating solids, the steps will hint on what gas is being formed as the instructions will clearly write what to prepare. For example, "prepare 3 cm depth of limewater....."

5. Record clear and concise observations. If possible, copy fully from the given notes for QA. Record correctly.

For example,

- (ppt ≠ solution)
- White precipitate formed vs no precipitate formed / no visible reaction.
  ~~White solution is seen.~~
- Purple acidified potassium manganate(VII) turns colourless
  ~~Yellow solution was seen.~~

6. Solids when heated, is mostly using strong flame (air hole fully opened).

Observe the difference between melt/sublime/residue colour change/droplets of colourless liquid.

7. When observing small changes in solution colours. Use original sample to compare if difficult to see.

8. When not heating, might want to touch test-tube lightly to check if exothermic or endothermic.

9. Conclude logically and **state the parts in the steps** that lead to the conclusion clearly. Remember to quote evidence (which part) when writing conclusion. If the evidence is not found in observations, it is not counted.

If you are confident the cations/anions of the unknown, you may try to see if everything make sense on your observations.

10. Read the question carefully. Name? Formulae?

11. When suggest **nature/property** : Try **oxidizing/reducing/acidic/alkaline/catalyst**

*Special questions :

Higher Valency - Iron(II) vs Iron(III), Which **part of periodic table** it is from : Alkali metals / Transition Metals/Halogens, Which **elements** does the substance have? Is it a **basic/acidic/amphoteric oxide?**

***Higher level Question:**

What is the difference between the following steps?

A. Add dilute acid* first followed by aqueous barium nitrate/barium chloride/silver nitrate.

Precipitate formed. → confirm means sulfate/chloride ions present!

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B. Add aqueous barium nitrate/barium chloride/silver nitrate first and see a precipitate.

Followed by addition of dilute acid, to the precipitate.

If precipitate remains undissolves in acid → means sulfate/chloride ions present!

precipitate dissolves in acid, producing bubbles of gases → it is carbonate ions!

All carbonates are insoluble, but they react with acids!

Given in EXAM!!

NOTES FOR QUALITATIVE ANALYSIS

Test for anions

anion	test	test result
carbonate (CO_3^{2-})	add dilute acid	effervescence, carbon dioxide produced
chloride (Cl^-) [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	white ppt.
iodide (I^-) [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	yellow ppt.
nitrate (NO_3^-) [in solution]	add aqueous sodium hydroxide, then aluminium foil; warm carefully	ammonia produced
sulfate (SO_4^{2-}) [in solution]	acidify with dilute nitric acid, then add aqueous barium nitrate	white ppt.

Test for aqueous cations

cation	effect of aqueous sodium hydroxide	effect of aqueous ammonia
aluminium (Al^{3+})	white ppt., soluble in excess giving a colourless solution	white ppt., insoluble in excess
ammonium (NH_4^+)	ammonia produced on warming	–
calcium (Ca^{2+})	white ppt., insoluble in excess	no ppt.
copper(II) (Cu^{2+})	light blue ppt., insoluble in excess	light blue ppt., soluble in excess giving a dark blue solution
iron(II) (Fe^{2+})	green ppt., insoluble in excess	green ppt., insoluble in excess
iron(III) (Fe^{3+})	red-brown ppt., insoluble in excess	red-brown ppt., insoluble in excess
zinc (Zn^{2+})	white ppt., soluble in excess giving a colourless solution	white ppt., soluble in excess giving a colourless solution

Test for gases

gas	test and test result
ammonia (NH_3)	turns damp red litmus paper blue
carbon dioxide (CO_2)	gives white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine (Cl_2)	bleaches damp litmus paper
hydrogen (H_2)	'pops' with a lighted splint
oxygen (O_2)	relights a glowing splint
sulfur dioxide (SO_2)	turns aqueous acidified potassium manganate(VII) from purple to colourless